A.C. CHARACTERISTICS AND WAVEFORMS

T_A = 0°C to 70°C, V_{CC} = 5V ±5%

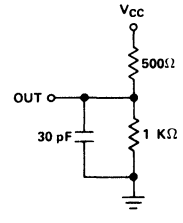
SYMBOL	PARAMETER	MIN	TYP(1)	MAX	UNIT
t_{CY}	Clock Cycle Time (2)	100	70		ns
t_{WP}	Clock Pulse Width	33	20		ns
t_{FS}	Function Input Set-Up Time (F_0 through F_6)	60	40		ns
	Data Set-Up Time:				
t_{DS}	I_0, I_1, M_0, M_1, K_0, K_1	50	30		ns
t_{SS}	LI, CI	27	13		ns
	Data and Function Hold Time:				
t_{FH}	F_0 through F_6	5	-2		ns
t_{DH}	I_0, I_1, M_0, M_1, K_0, K_1	5	-4		ns
t_{SH}	LI, CI	15	2		ns
	Propagation Delay to X, Y, RO from:				
t_{XF}	Any Function Input		37	52	ns
t_{XD}	Any Data Input		29	42	ns
t_{XT}	Trailing Edge of CLK		40	60	ns
t_{XL}	Leading Edge of CLK	20			ns
	Propagation Delay to CO from:				
t_{CL}	Leading Edge of CLK	20			ns
t_{CT}	Trailing Edge of CLK		48	70	ns
t_{CF}	Any Function Input		43	65	ns
t_{CD}	Any Data Input		30	55	ns
t_{CC}	CI (Ripple Carry)		14	25	ns
	Propagation Delay to A_0, A_1, D_0, D_1 from:				
t_{DL}	Leading Edge of CLK	5	32	50	ns
t_{DE}	Enable Input ED, EA		12	25	ns

NOTE:
(1) Typical values are for T_A = 25°C and nominal supply voltage.
(2) $t_{CY} = t_{DS} + t_{DL}$.

TEST CONDITIONS:

Input pulse amplitude: 2.5 V
Input rise and fall times of 5 ns between 1 and 2 volts.
Output loading is 10 mA and 30 pF.
Speed measurements are made at 1.5 volt levels.

TEST LOAD CIRCUIT:



CAPACITANCE(2) T_A = 25°C

SYMBOL	PARAMETER	MIN	TYP	MAX	UNIT
C_{IN}	Input Capacitance		5	10	pF
C_{OUT}	Output Capacitance		6	12	pF

NOTE:
(2) This parameter is periodically sampled and is not 100% tested. Condition of measurement is f = 1 MHz, V_{BIAS} = 2.5V, V_{CC} = 5.0V and T_A = 25°C.